Literature Review: Deep Learning for Seizure Detection and Prediction from Non-EEG Wearable Signals (2013–2026)

Paulin Saher
Universität zu Köln

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Contents

Abstract

Background

Epilepsy affects approximately 60 million people worldwide. One third of patients remain drug-resistant, and 69% of deaths from generalized tonic-clonic seizures could be prevented with timely detection (Sveinsson et al., 2020). EEG-based prediction is effective but unsuitable for ambulatory use due to operational complexity. Wearable devices using autonomic and motion signals offer an alternative, but translating deep learning approaches from controlled settings to real-world deployment remains challenging.

Methods

Following PRISMA guidelines and the Webster & Watson concept matrix framework, this systematic review synthesizes 13 primary studies published between 2013 and 2026. Nine studies address detection and four address forecasting. Studies were evaluated based on algorithmic architecture, biosignal modality, validation methodology, sensitivity, and false alarm rate.

Results

Convulsive seizure detection achieves high sensitivity (71.6-100%) primarily through accelerometry. Only two systems achieve the ILAE Phase 3 FPR benchmark of 0.008-0.028/h with the caviat that they were only validated in EMU: Spahr et al. (2025) (96% sensitivity, 0.0054/h) and Fine (2025) (100% sensitivity, 0.023/h). ECG-based detection fails clinically due to excessive false alarm rates ranging from 0.11 to 65.62/h (Reintjes et al., 2025).
Seizure forecasting faces a fundamental performance ceiling. Across methods and modalities, AUC consistently plateaus around 0.74-0.75 (Meisel et al., 2020; Vieluf et al., 2025). Under rigorous leave-one-subject-out validation, only 43% of patients achieve better-than-chance performance. Patient-specific models achieve 100% success but require weeks to months of individual data, creating a cold-start problem for newly diagnosed patients.
Multi-modal approaches dominate research (69% of studies) yet the best detection result uses single-modality accelerometry, suggesting biological specificity matters more than sensor count. Wrist-worn devices predominate (62% of studies). Validation rigor is insufficient: only 3 of 13 studies employ prospective designs, and 11 of 13 validate in hospital settings that likely underestimate real-world false alarm rates.

Conclusions

Wearable seizure detection has almost reached clinical viability for nocturnal convulsive seizures with a translation gap from EMU to home. Forecasting remains immature without established benchmarks or regulatory pathways. Three gaps impede progress: lack of large-scale home validation, absence of standardized forecasting benchmarks, and no

viable solutions for approximately 81% of patients with focal onset seizures (Ayub et al., 2020). False alarm rates remain the primary barrier for daytime detection. Future research must prioritize prospective validation in intended use environments, address the cold-start problem through adaptive personalization, and identify novel biomarkers for focal seizures without motor manifestations.

1 Introduction

1.1 Problem Statement and Motivation

Epilepsy, a chronic neurological disorder affecting 60 million people worldwide, with approximately one-third of patients remaining drug-resistant (Hussein et al., 2018). Sveinsson et al. (2020) found that 69% of deaths due to a generalized tonic-clonic seizure could have been prevented if not left unattended, highlighting the demand for reliable and timely prediction.

While reliable prediction is already possible through electroencephalography (EEG), its operational complexity renders it unsuitable for ambulatory use. To address the need for ambulatory detection many are experimenting with utilizing autonomic and motion signals to detect and predict seizures.

Translating and analysing these often noisy and complex time-series physiological signals has pushed traditional machine learning (ML) approaches to their limits. These approaches require manual engineering such as cleaning and extracting of the data, often failing to capture the complex, non-linear temporal dynamics of these signals. It is this limitation that deep learning (DL) models are designed to overcome, with their ability of hierarchical and learned feature extraction uniquely addressing these limitations.

This review therefore examines and assesses recent advances in time-series deep learning - particularly convolutional neural networks (CNNs) and long short term memory (LSTMs) - applied to seizure detection and prediction using non-EEG wearable biosignals. Central to this analysis is the question of how architectural choices, training paradigms (e.g., transfer learning, personalization), and thorough prospective evaluation can converge to meet the stringent clinical requirements of high sensitivity and minimal false alarm rates (FAR) in real-world deployment.

1.2 Key Research Question

How do different deep learning architectures and biosignal modalities excluding EEG compare in their ability to achieve an optimal trade-off between sensitivity and false alarm rate for ambulatory seizure monitoring?

1.3 Clinical Evaluation Frameworks

1.3.1 ILAE Phased Validation Framework

The clinical validity of seizure detection algorithms is assessed through the International League Against Epilepsy (ILAE) phased framework, which classifies studies into five validation phases (Beniczky & Ryvlin, 2018; Beniczky et al., 2021).

Phase 0 represents proof of concept. Phase 1 involves retrospective analysis with small samples. Phase 2 requires minimum 10 patients with seizures and 15 recorded seizures.

Phase 3 denotes prospective, multicenter validation with at least 30 seizures from at least 20 patients using real-time detection. Phase 4 represents real-world home validation.
Prospective trials (Phase 3) constitute the clinical gold standard, requiring locked algorithms, video-EEG reference standards, and testing on unseen data from multiple centers.

1.3.2 Study Design Classifications

This review distinguishes three study design types. Retrospective analysis examines historical data and risks substantial data leakage when non-chronological splits allow future information to inform training (Kalousios et al., 2024; Wong et al., 2023).
Pseudo-prospective validation enforces chronological order, training only on past seizures to test on subsequent ones, providing more realistic performance estimates (Kalousios et al., 2024; Lu et al., 2025).
Prospective studies employ real-time testing on new patients with locked algorithms, representing the highest validation rigor.

1.3.3 Clinical Performance Benchmarks

The ILAE systematic review established performance benchmarks for generalized tonic-clonic seizures (GTCS, bilateral motor seizures affecting both sides of the brain with loss of consciousness) and focal-to-bilateral tonic-clonic seizures (FBTCS, seizures starting in one hemisphere and spreading to both). Validated devices achieve 90-96% sensitivity with false alarm rates of 0.2-0.67 per 24 hours (approximately 0.008-0.028/h) (Beniczky et al., 2021).
For ambulatory use, acceptable false alarm rates depend on user perspective. Patients and caregivers typically require 100% sensitivity and accept approximately one false alarm per week. Clinicians consider 90% sensitivity adequate and tolerate two false alarms per week to one per month (Sivathamboo et al., 2022).
The ILAE Phase 3 benchmark of 0.008-0.028/h represents the clinically validated performance standard for GTCS/FBTCS detection.
Currently, the ILAE recommends wearable devices only for GTCS/FBTCS detection in unsupervised patients where alarms can enable rapid intervention. This represents a weak recommendation based on moderate evidence. For all other seizure types, the ILAE does not recommend clinical use of currently available wearable devices (Beniczky et al., 2021).

2 Methodological Approach and Structure

The review follows structured IS/medical literature guidance (concept matrix inspired by (Webster & Watson, 2002) and PRISMA principles (Tugwell & Tovey, 2021)). The methodological approach uses a multi-stage search and transparent extraction and synthesis procedures as outlined below.

1. **Search strategy (Multi-stage, 2015-2025):**
    - **Scoping:** Google Scholar for broad coverage and identifying candidate papers.
    - **Formal queries:** IEEE Xplore, PubMed, and Scopus using controlled keyword strings and deduplication.
    - **Citation chasing:** Backward and forward citation tracking on key papers to ensure completeness.

2. **Inclusion / Exclusion criteria:**
    - **Inclusion:** Studies using wearable/non-EEG autonomic signals for seizure detection or preictal prediction.
    - **Exclusion:** EEG-only studies, invasive recordings, or papers limited to purely algorithmic simulations without human data.

Extracted dimensions populate two concept matrix tables with the following fields:

1. **Study Matrix (Table 1 for detection, Table 2 for forecasting):**
    - Study identification (author, year)
    - Objective (detection vs forecasting, seizure type)
    - Design (prospective, retrospective, pseudo-prospective, LOSO CV)
    - Sample (participant count, seizure count)
    - Device and sensor location
    - Algorithm architecture
    - Performance metrics (sensitivity, specificity, FPR)
    - Key limitations
2. **Detailed Metrics Matrix (Table 3 for detection, Table 4 for forecasting):**
    - Validation approach (independent test, cross-validation, LOSO)
    - Detection latency
    - Patient-level success rate
    - Precision/PPV
    - Additional clinical metrics (HMS, IoC, AUC, F1)
    - Clinical notes

Additional Extracted Dimensions: The following dimensions were extracted and analyzed to inform the narrative synthesis and dimensional analysis presented in Section 4, complementing the tabular comparison:

- Setting (clinical vs ambulatory)
- Modality details (ECG/HRV, PPG, EDA, ACC)
- Evaluation granularity (sample- vs alarm-based)
- Windowing specifics (window length, stride, overlap)
- Personalization strategies
- Interpretability and safety considerations
- Data characteristics and dataset identity
- Missing data handling (imputation, exclusion) and synthetic data use

- Inference requirements (latency, computational load, hardware)

This two-table approach follows Webster & Watson's concept matrix methodology (Webster & Watson, 2002), balancing complete data extraction with readable summary tables while enabling detailed dimensional analysis in the narrative synthesis.
Prioritization Criteria: Studies including splits preserving patient-specific data, FAR/h reporting and external or cross validation will be prioritized in the synthesis. A PRISMA-style flow diagram will summarize and visualize screening and inclusion.

3 Search Strategy

3.1 PICO-Based Search Methodology

The literature search was structured using the PICO framework (Patient/Problem, Intervention, Comparison, Outcome) to identify relevant studies on deep learning for non-EEG epilepsy detection. The following key terms were used in combination with Boolean operators (AND between P,I,C,O, OR in the categories themselves and NOT):

1. **P (Patient/Problem):** Epileptic seizures in need of detection or prediction.
    - **Keywords:** 'epilepsy', 'seizure', 'ictal', 'preictal'
    - **MeSH Terms (PubMed):** Epilepsy, Seizures
2. **I (Intervention):** The diagnostic method based on wearable biosignals and deep learning.
    - **Biosignal Keywords:** 'electrocardiogram', 'ECG', 'heart rate variability', 'HRV', 'photoplethysmography', 'PPG', 'electrodermal activity', 'EDA', 'accelerometer', 'ACC'
    - **Technology Keywords:** 'wearable', 'wearable device'
    - **Method Keywords:** 'deep learning', 'machine learning', 'convolutional neural network', 'CNN', 'long short-term memory', 'LSTM', 'neural network'
    - **MeSH Terms (PubMed):** Wearable Electronic Devices, Electrocardiography, Deep Learning
3. **C (Comparison):** The explicit exclusion of studies focused solely on electroencephalography (EEG) as the primary modality.
    - **Exclusion Keywords:** 'EEG', 'electroencephalography', 'iEEG', 'intracranial'
    - **Search Logic:** The `NOT` operator was applied to this group in all database searches.
4. **O (Outcome):** Metrics quantifying the performance of a detection or prediction system.
    - **Keywords:** 'detection', 'prediction', 'sensitivity', 'specificity', 'false alarm rate', 'FAR', 'area under the curve', 'AUC'

3.2 Practical Filters and Reproducibility

Consistent with the overall review scope, searches were restricted to **2015-2025** and **journal articles** with few exceptions made regarding **conference proceedings**. All database searches were documented (query string, platform, and execution date). Results were deduplicated prior to screening.

Note on exclusions: Because some non-EEG wearable studies rely on video-EEG for labeling, EEG-related terms were used to filter **EEG-only modalities**, not to exclude studies that mention EEG solely as a reference standard.

3.3 Exploration

Due to the insufficient amount of papers gathered from systematic review an exploration was done on the already existing papers. The exploration consisted of graph tools to visualize backlinks from the 9 already found studies and gemini deep research feature to find relevant papers. Through this method an additional 4 papers were aquired bringing the total number of papers up to 13 and expanding the time frame to 2013-2026.

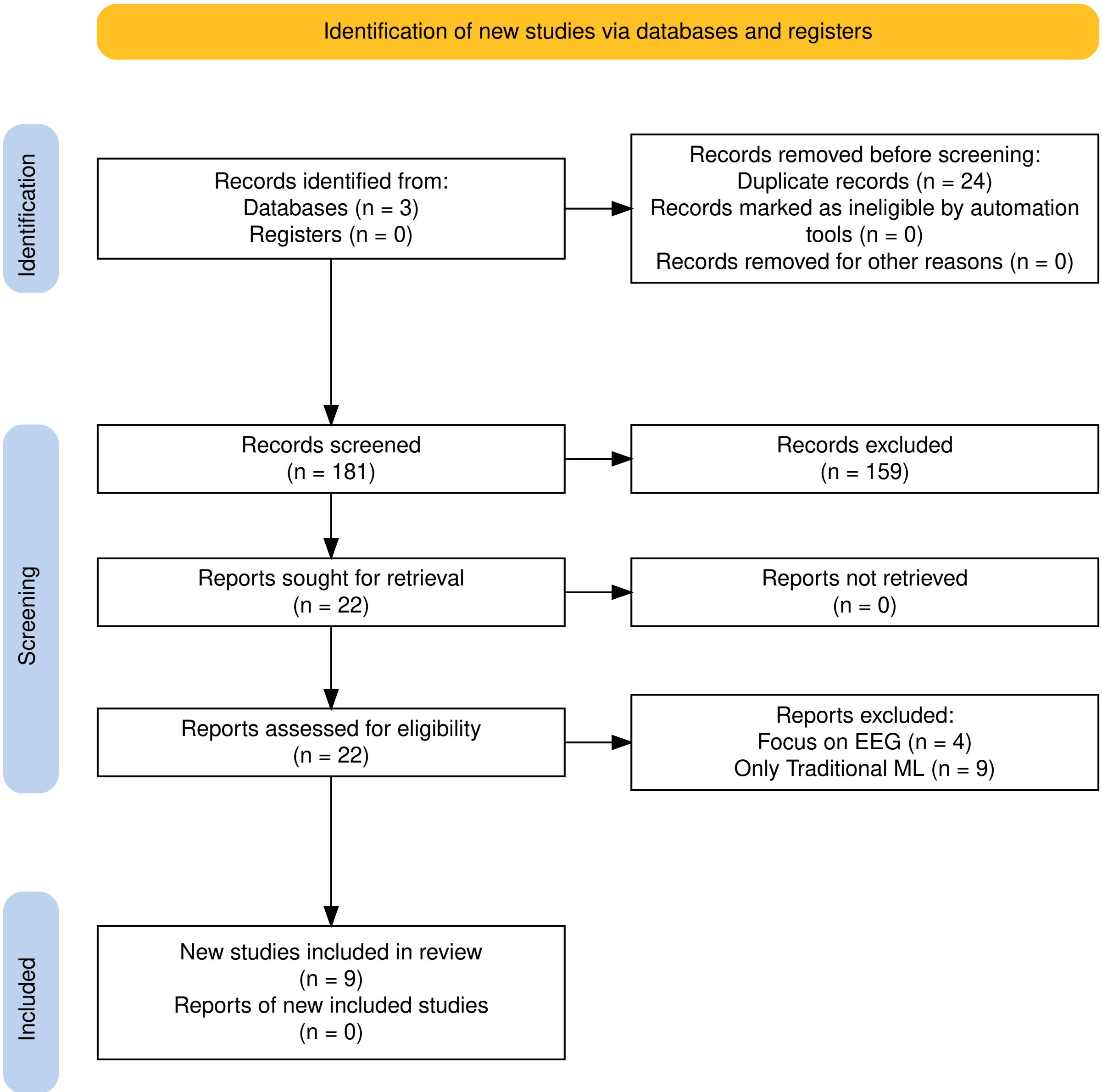

Figure 1: PRISMA flow diagram of the study selection process.

Table 1: Overview: Detection Studies

Study	Design	Sample	Device	Loc.	Para.	Mod.	Pers.	Algorithm	Sens	Spec	FPR
Spahr et al. 2025	Prospective	n=384, 49 CSs	Empatica E4 (ACC)	Wrist	Ens	1	Glb	Ensemble 1D CNN (30 models)	96%	–	0.0054/h
Reintjes et al. 2025	Retro	n=120, 856 szs	Single-lead ECG	Chest	Anom	1	Subj	Anomaly: Matrix Profile, MADRID, TimeVQVAE	2.44–82.79%	–	0.11–65.62/h
Fine 2025	Phase 1	n=15/3	6-axis band (ACC+Gyro)	Wrist	Feat	M	Glb	ANN (594 handcrafted feat)	100%Test, 96% complete	–	0.023/h
Dong et al. 2026	Prospective	n=68, 1846 szs	NightWatch	Arm	2-stg	M	Glb	CNN-LSTM + Attention	71.6%	–	0.165/h
Wang et al. 2025	–	n=28, 62 szs	Biovital-P1	Wrist	Feat	M	Glb	LSTM (40 hidden)	56.40% to 95.3%	–	0.364/h
Singh Rathore 2024	Case study	1 pt, 6 hrs	Wearable (EDA+ACC+HR)	–	Feat	M	CS	MLP (multimodal features)	96.8%	94.8%^{prec}	–
Elemam et al. 2025	Cross-sect	Q:198, HRV:30	Camera PPG	Thumbs	Hyb	M	Glb	CNN + Rule-based fusion	95.1% Q, 93% HRV	97.06% Q	–
Borujeny 2013	3-pt	3 pts, 20 szs	MICAz ACC	Arm/thigh	Feat	1	Glb	ANN (time-domain feat)	85%	–	3 FA

Note: ACC = accelerometer, ANN = artificial neural network, CNN = convolutional neural network, CS = convulsive seizure / case study, ECG = electrocardiogram, EDA = electrodermal activity, HR = heart rate, HRV = HR variability, LSTM = long short-term memory, PPG = photoplethysmography, Sens = sensitivity, Spec = specificity, sz(s) = seizure(s), FPR = false positive rate, – = not reported. Para. = learning paradigm (Feat = feature-based, Ens = ensemble, Anom = anomaly detection, 2-stg = two-stage, Hyb = hybrid). Mod. = modality count (1 = single-modal, M = multi-modal). Pers. = personalization (Glb = global model, Subj = subject split, CS = case study). ^{prec} = precision (not specificity).

Table 2: Overview: Forecasting Studies

Study	**Design**	**Sample**	**Device**	**Loc.**	**Para.**	**Mod.**	**Pers.**	**Algorithm**	**Sens**	**Spec**	**FPR**
Vieluf et al. 2025	Retro	n=70, 5437 d	Embrace	Wrist	Hyb	M	Mix	DNN + harmonic features + diary	82%	67%	–
Meisel et al. 2020	LOSO CV	n=69, 452 szs	Empatica E4	Wrist/Ankle	E2E	M	Glb	LSTM (10 units)	51.2%	–	TiW 43.7%
Stirling 2021	Retro+pseudo	n=11, 136 szs	Fitbit	Wrist	Feat	M	Pat	LSTM+RF+LR ensemble (HRV feat)	–	–	AUC 0.74
Nasseri 2021	Retro	n=6	Empatica E4	Wrist	E2E	M	Tmp	LSTM 4-layer (128 hidden)	AUC 0.75 (SD 0.15)	–	TiW 0.9–7.2h/d
Ode et al. 2023	Retro	n=66, 85 szs	ECG	–	Anom	1	Pat	Self-Attentive AE	74%	–	0.85/h

Note: ACC = accelerometer, AE = autoencoder, CNN = convolutional neural network, DNN = deep neural network, ECG = electrocardiogram, EDA = electrodermal activity, LSTM = long short-term memory, LOSO = leave-one-subject-out, PPG = photoplethysmography, Sens = sensitivity, Spec = specificity, sz(s) = seizure(s), FPR = false positive rate, TiW = time in warning, AUC = area under curve, – = not reported. Para. = learning paradigm (Feat = feature-based, E2E = end-to-end, Anom = anomaly detection, Hyb = hybrid). Mod. = modality count (1 = single-modal, M = multi-modal). Pers. = personalization (Glb = global model, Pat = patient-specific, Mix = mixed, Tmp = temporal split).

Table 3: Architecture and Modality: Detection Studies

Study	Paradigm	Arch. Details	Modality Details	Fusion	Personalization	Trade-off Config	Deployment
Spahr et al. 2025	Ensemble	30 1D CNN models, 14 conv layers, quantile	ACC 32 Hz, Euclidean norm, 30 s	–	Global, tunable	86–100% @ 0.0054/h	Real-time (112 ms), on-device
Reintjes et al. 2025	Anomaly	Matrix-Profile, MADRID, TimeVQVAE-AD	ECG 256→8 Hz, bandpass	–	Subject split	2.44–82.79% @ 0.11–65.62/h	Offline, hospital
Fine 2025	Feature-based	ANN, 594 handcrafted features	ACC+Gyro 6-axis, 10 s intervals	Early	Global, hold-out	100% @ 0.023/h	Offline PC, EMU
Dong et al. 2026	Two-stage	Pre-screening + CNN-LSTM + Attention	ACM 11-12→20 Hz, PPG 100→20 Hz, 5 min	Early	Global, 10-fold CV	71.6% @ 0.165/h	Real-time, NightWatch, home
Wang et al. 2025	Feature-based	LSTM (40 hidden) + ReLU + FC	ACC/GYR 50 Hz, SEMG 200 Hz, EDA 4 Hz, 4 s	Early	Global, hold-out	56.4–95.3% @ 0.364/h	Real-time, hospital
Singh 2024	Feature-based	MLP, multi-modal features	EDA, ACC, HR/HRV, 25k points	Early	Single pt (CS)	96.8% @ 94.8% prec	Real-time, cloud?
Elemam 2025	CNN-based	CNN for HRV, CNN for Audio (separate)	PPG 250 Hz, Audio 10 s	No fusion	Global, unclear	95.1% @ 97.06% spec	Real-time, hospital
Borujeny 2013	Feature-based	ANN, time-domain features	ACC 2D 3 Hz, 9 s	–	Global, 3 pts	85% @ 3 FA	Real-time, server (316 mW)

Note: Paradigm: Feature-based = handcrafted features, Ensemble = multiple models, Two-stage = sequential processing, Anomaly = unsupervised detection, CNN-based = convolutional approach. Arch. Details: architecture-specific components. Fusion: – = single-modal, Early = feature-level, No fusion = parallel separate models. Personalization: Global = population model, Subject split = train/test separated by subject. Trade-off Config: operating points with sensitivity @ FAR. Deployment: real-time capability, processing location, setting. ACC = accelerometer, ECG = electrocardiogram, PPG = photoplethysmography, EDA = electrodermal activity, HR = heart rate, HRV = HR variability, SEMG = surface electromyography, EMU = epilepsy monitoring unit.

Table 4: Architecture and Modality: Forecasting Studies

Study	Paradigm	Arch. Details	Modality Details	Fusion	Personalization	Trade-off Config	Deployment
Vieluf et al. 2025	Hybrid	DNN + harmonic modeling (24-h)	EDA 4 Hz, ACC 32 Hz, Temp 1 Hz, 10 min	Early	Mixed, 5-fold + LOO	82% sens, 67% spec @ F1 0.81	Offline MATLAB, home
Meisel et al. 2020	End-to-end	LSTM only (10 units)	EDA/ACC/BVP/Temp →4 Hz, 30 s	Early	Global, LOSO	51.2% @ IoC 14.1%	Offline, hospital
Stirling 2021	Feature-based	LSTM+RF+LR ensemble, LR combines	HR 5 s, steps/sleep 1 min, hourly/daily	Decision	Patient-specific, weekly	AUC 0.74, 100% pts	Home (Fitbit), app
Nasseri 2021	End-to-end	LSTM 4-layer, 128 hidden + FFT channels	ACC/BVP/EDA/Temp/HR →128 Hz, 60 s	Early (17-ch)	Temporal split	AUC 0.75 (SD 0.15) @ TiW 0.9–7.2h/d	Offline, cloud, home
Ode et al. 2023	Anomaly	Self-Attentive AE (attention)	ECG (RRI only), 45 s	–	Patient-specific (99% CL)	74% @ 0.85/h (99% CL)	Real-time, cloud, hospital

Note: Paradigm: Feature-based = handcrafted features, End-to-end = learns from raw data, Hybrid = combines both, Anomaly = unsupervised detection. Arch. Details: architecture-specific components. Fusion: – = single-modal, Early = feature-level, Decision = classifier output. Personalization: Global = population model, Patient-specific = individualized, Temporal = time-split within patient, LOSO = leave-one-subject-out, LOO = leave-one-out, Mixed = combination of approaches. Trade-off Config: operating points with sensitivity @ FAR. Deployment: real-time capability, processing location, setting. ACC = accelerometer, ECG = electrocardiogram, EDA = electrodermal activity, BVP = blood volume pulse, HR = heart rate, TEMP = temperature, FFT = fast Fourier transform, CL = confidence limit, AUC = area under curve, TiW = time in warning, IoC = Improvement over Chance.

Table 5: Performance Metrics Reporting Across Studies

Metric	Detection (n=9)	Forecasting (n=4)	Reported By
Sensitivity/Recall	8/8	1/4	Spahr, Reintjes, Fine, Dong, Wang, Singh Rathore, Elemam, Borujeny, Ode[a]
Specificity	3/9	0/4	Elemam, Vieluf, Wang
FPR/FA rate	8/9	0/4	Spahr, Reintjes, Fine, Dong, Wang, Singh Rathore, Elemam, Borujeny, Nasseri, Ode
AUC-ROC	3/9	3/4	Dong, Reintjes, Ode, Nasseri, Stirling
Detection Latency	4/9	1/4	Spahr, Fine, Elemam, Borujeny, Nasseri
Patient Success Rate	2/9	3/4	Reintjes, Meisel, Stirling, Nasseri, Ode
Precision/PPV	4/9	1/4	Dong, Singh Rathore, Elemam, Ode
IoC / Time in Warning	0/9	3/4	Meisel, Stirling, Nasseri

4 Results

This systematic review analyzed 13 primary studies published between 2013 and 2026. The evidence base comprises 9 detection studies and 4 forecasting studies with a total of 912 participants. Table 1 and Table 2 present the study matrices for detection and forecasting approaches, respectively. Table 3 and Table 4 provide architectural details, and Table 5 summarizes the metrics reported across studies.

4.1 Study Characteristics

The 13 studies included in this review exhibit substantial variation in sample size and study design. Detection studies enrolled a median of 44 participants (range 1 to 384), while forecasting studies enrolled a median of 11 participants (range 6 to 70). The total sample comprises 687 participants in detection studies and 225 in forecasting studies.

Three studies employed prospective designs. Spahr et al. (2025) conducted a multi-center prospective study across eight epilepsy monitoring units. Dong et al. (2022) prospectively monitored patients at home for up to three months. Fine (2025) prospectively recorded data from 15 patients in an epilepsy monitoring unit for tonic seizure detection. The remaining studies used retrospective analysis of existing datasets or small-scale feasibility designs.

Research activity has accelerated over time. Only one study was published between 2013 and 2015. Three forecasting studies were published between 2020 and 2022. Nine detection studies were published between 2023 and 2026. This pattern suggests increased research attention on detection approaches in recent years.

Eight studies were conducted in hospital or EMU settings. Six studies included home or ambulatory validation.

Key limitation. The evidence base includes four studies with sample sizes below 20 participants. Singh Rathore et al. (2024) is a single-patient case study with limited generalizability. Borujeny et al. (2013) evaluated only 3 patients. Small sample sizes limit the reliability of performance estimates and generalizability to broader populations.

4.2 Modality Performance

Biosignal modality selection differs substantially between detection and forecasting approaches. Detection studies prioritize movement sensors, while forecasting studies emphasize autonomic nervous system indicators.

4.2.1 Modality Usage

Accelerometer data appears in nine of 13 studies (69%). Six detection studies and three forecasting studies use accelerometry. Electrodermal activity appears in five studies (38%). ECG or HRV data appears in five studies. PPG appears in four studies. Gyroscope data appears in three studies, all in detection applications.

Multi-modal approaches dominate the evidence base. Nine of 13 studies (69%) use multiple sensor types. Four studies use single-modality approaches: Spahr et al. (2025) (ACC only), Reintjes et al. (2025) (ECG only), Borujeny et al. (2013) (ACC only), and Ode et al. (2023) (ECG RRI only).

4.2.2 Sensor Location

Wrist-worn devices appear in eight studies (62%). Arm or armband placement appears in three studies. Chest placement appears in one study. Thigh placement appears in one study. Borujeny et al. (2013) used multiple sensor locations (right arm, left arm, left thigh).

The preference for wrist placement reflects patient acceptance and the availability of consumer wearable devices. The Empatica E4 wristband appears in three studies. The Embrace watch appears in one study. Fitbit smartwatches appear in one study.

4.2.3 Detection Performance by Modality

Single-modality accelerometer approaches achieve strong performance. Spahr et al. (2025) reported 96% sensitivity with 0.0054/h FPR using only ACC data from the Empatica E4. This represents the lowest FPR among all detection studies. Borujeny et al. (2013) reported 85% sensitivity using 2D ACC, though with only three patients.

ECG-based detection is currently clinically non-viable due to unacceptable false alarm rates. Reintjes et al. (2025) compared three anomaly detection methods on single-lead ECG. Sensitivity ranged from 13.4% to 82.8%, while FPR ranged from 0.11/h to 65.62/h depending on method and configuration. The lowest FPR of 0.11/h detected only 2.44% of seizures. Ode et al. (2023) achieved 74% sensitivity with 0.85/h FPR using ECG RRI data.

Wang et al. (2025) directly compared modality performance. Attitude angle features (PITCH and ROLL) outperformed raw accelerometer and gyroscope data for seizure detection. The architecture achieved 56.4% to 95.3% sensitivity depending on the feature configuration. This finding suggests that feature engineering can enhance the value of standard movement sensors.

4.2.4 Forecasting Performance by Modality

Forecasting studies consistently use autonomic signals. Nasseri et al. (2021) combined ACC, PPG, EDA, temperature, and HR to achieve AUC 0.75 (SD 0.15). Meisel et al. (2020) used EDA, PPG, temperature, and ACC from the Empatica E4 to achieve 51.2% sensitivity with IoC 14.1% in LOSO validation.

Vieluf et al. (2025) combined EDA, ACC, and temperature with diary data. This study demonstrated that 24-hour harmonic patterns in EDA and accelerometry contain predictive information.

Key finding. Multi-modal approaches are common but not universally superior. Single-modality ACC (Spahr et al. (2025)) achieves the best FPR in the included detection literature. Forecasting consistently requires multi-modal autonomic data, but AUC-ROC plateaus around 0.75 regardless of modality combination.

4.3 Architectures and Personalization

4.3.1 Learning Paradigms

Deep learning approaches in wearable seizure monitoring employ diverse learning paradigms. Feature-based approaches appear in four studies. These methods extract handcrafted features as input to classifiers. Fine (2025) extracted 594 features from 6-axis ACC and gyroscope data for an ANN classifier. Wang et al. (2025) engineered attitude angle features for an LSTM network with 40 hidden units. Singh Rathore et al. (2024) used multi-modal features for an MLP classifier in a single-patient case study.

End-to-end learning appears in two studies, both in forecasting. Meisel et al. (2020) used an LSTM with 10 units operating directly on raw multi-modal wrist data. Nasseri et al. (2021) used a 4-layer LSTM with 128 hidden units per layer receiving FFT channels of raw sensor data.

Hybrid approaches appear in two studies. Vieluf et al. (2025) combined a DNN with handcrafted harmonic features extracted from 24-hour patterns. Elemam et al. (2025) used separate CNNs for HRV and audio data without fusion between modalities.

Ensemble methods appear in detection applications. Spahr et al. (2025) trained 30 separate 1D CNN models with quantile aggregation, achieving 96% sensitivity with 0.0054/h FPR. The ensemble aggregation quantile serves as a tunable parameter that allows adapting the sensitivity-FPR trade-off according to patient needs. The model (Episave) operates on 3D-ACC data at 32 Hz using 30-second windows. Dong et al. (2022) used a two-stage architecture with threshold-based pre-screening followed by a CNN-LSTM model.

Anomaly detection appears in two studies. Reintjes et al. (2025) compared three unsupervised anomaly detection methods (Matrix Profile, MADRID, TimeVQVAE) on single-lead ECG data. Ode et al. (2023) used a self-attentive autoencoder for patient-specific anomaly detection on ECG RRI intervals.

4.3.2 Architecture Patterns by Application

Deep learning architectures differ substantially between detection and forecasting applications. Forecasting studies concentrate on LSTM-based approaches while detection research is more diverse but CNN-based approaches dominate.

CNN-based methods are prominent in detection. Spahr et al. (2025) implemented an ensemble of 30 CNN models with 14 convolutional layers each, operating on ACC data sampled at 32 Hz with 30-second windows. Elemam et al. (2025) used separate CNNs for HRV and audio processing. Traditional neural networks appear in earlier studies, with Fine (2025) using an ANN with 594 handcrafted features achieving 100% sensitivity.

Three of five forecasting studies use LSTM variants. Meisel et al. (2020) implemented a minimal LSTM with 10 hidden units operating directly on raw multi-modal wrist data. Nasseri et al. (2021) used a deeper 4-layer LSTM with 128 hidden units per layer. Stirling et al. (2021) combined LSTM with random forest and logistic regression in an ensemble. No forecasting study uses CNN-based approaches, possibly because forecasting requires modeling temporal dependencies rather than spatial patterns.

Handcrafted features remain competitive. Three of eight detection studies use feature-based approaches with strong performance. Fine (2025) achieved 100% sensitivity using 594 handcrafted features. This suggests that domain knowledge remains valuable despite the end-to-end learning trend.
Window sizes vary substantially between applications. Detection studies use shorter windows ranging from 4 seconds to 5 minutes. Forecasting studies use longer windows ranging from 30 seconds to 60 minutes.

4.3.3 Personalization Strategies

Personalization strategy represents a fundamental design choice. Global models train on population data and apply to all patients while patient-specific models train on individual patient data. Five studies use global model approaches, four studies use patient-specific approaches, and one study uses a mixed approach.

Global Models. Five studies use global model approaches. Spahr et al. (2025) trained a global model on 384 patients with tunable parameters. Fine (2025) trained on 15 patients and tested on 3 patients. Dong et al. (2022) trained on 68 patients with 10-fold cross-validation. Elemam et al. (2025) trained on 198 questionnaire responses and 30 HRV recordings. Meisel et al. (2020) trained a global model tested with LOSO validation.
Global models enable on-device processing without requiring individual patient training. Spahr et al. (2025) achieved 112 ms inference time suitable for real-time processing. However, global models may not capture individual seizure patterns. Meisel et al. (2020) used LOSO validation to assess global model generalizability, achieving 62% patient success (43 of 69 patients above chance).

Patient-Specific Models. Four studies use patient-specific approaches. Stirling et al. (2021) trained individual models with weekly retraining for 11 patients. Nasseri et al. (2021) trained separate models for each of 6 patients. Ode et al. (2023) trained anomaly detection models at the 99% confidence level for each of 66 patients. Singh Rathore et al. (2024) presented a single-patient case study.
Patient-specific models achieve higher individual success rates. Stirling et al. (2021) achieved 100% patient success for hourly forecasting. Nasseri et al. (2021) achieved 83% patient success (5 of 6 patients). These rates are substantially higher than the 62% achieved by global models with LOSO validation. However, patient-specific approaches require a mean of 14.6 months of data per patient or 60+ days per patient. This data requirement creates a cold start problem where new patients must wait weeks or months before the system becomes useful.

Mixed Approaches. Vieluf et al. (2025) used a mixed approach. The study combined population-level 24-hour harmonic patterns with individual diary data. This hybrid achieved 82% sensitivity and 67% specificity. The approach may balance the advantages of global and patient-specific methods.

4.3.4 Validation Approaches

Validation approaches vary across studies. Reintjes et al. (2025) used subject-split validation, training on one set of patients and testing on different patients to assess generalization. The results showed poor generalization with FPR ranging from 0.11/h to 65.62/h depending on method and configuration.

Nasseri et al. (2021) used temporal split validation. Early data from each patient served as training while later data served as testing. This approach assesses temporal stability rather than generalization to new patients.

Meisel et al. (2020) used LOSO validation to assess global model performance on unseen patients, achieving 62% patient success.

4.3.5 Temporal Evolution

Architecture choices have evolved over time. The earliest study, Borujeny et al. (2013), used traditional ANN approaches on MICAz accelerometer data in a 3-patient study. Studies published between 2020 and 2022 focused on LSTM-based forecasting architectures. Studies published between 2023 and 2026 employed more complex architectures including ensembles, hybrid methods, and attention mechanisms.

4.3.6 Computational Considerations

Global models enable on-device deployment. Spahr et al. (2025) achieved 112 ms inference time suitable for real-time processing. On-device processing enables continuous monitoring without cloud connectivity.

Patient-specific models often require cloud or smartphone-based processing. Stirling et al. (2021) used smartphone-based processing with weekly retraining. Nasseri et al. (2021) used cloud-based processing. Cloud-based approaches may experience connectivity issues and latency.

Key findings. Detection research employs architectural diversity including CNNs, LSTMs, ANNs, and anomaly detection. Forecasting research concentrates heavily on LSTM variants with no CNN-based approaches, possibly because forecasting requires modeling temporal dependencies rather than spatial patterns. Patient-specific models achieve higher individual success (83-100%) compared to global models with LOSO validation (62%), but they require weeks to months of individual data. Global models enable immediate deployment and on-device processing but may not work well for all patients.

Limitations. Architecture descriptions vary in detail across studies. Some studies report complete architectural details while others provide minimal information. This variability complicates direct comparison and replication. Detection studies rarely report patient-level success. Only two of eight detection studies provide individual patient results. This reporting gap prevents assessment of which patients benefit from global detection models.

4.4 Performance Metrics

Detection and forecasting studies use different evaluation metrics reflecting their distinct clinical objectives. Detection studies emphasize sensitivity and false positive rate (FPR). Forecasting studies report AUC-ROC, improvement over chance (IoC), and time in warning (TiW).

4.4.1 Detection Performance

Detection sensitivity ranges from 38.0% to 100% across studies (Table 1). Individual study results are reported in Sections 4.2 and 4.3. Notable high-performing studies include Fine

(2025) at 100% sensitivity, Spahr et al. (2025) at 96% sensitivity, Singh Rathore et al. (2024) at 96.8% sensitivity in a single-patient case study, and Elemam et al. (2025) at 95.1% sensitivity using HRV-based detection.
Only six of nine detection studies report FPR. The ILAE systematic review of Phase 3 studies reported FPR of 0.2-0.67 per 24 hours (approximately 0.008-0.028/h) for validated devices (Beniczky et al., 2021). Two studies meet this performance range. Spahr et al. (2025) achieved 0.0054/h FPR (approximately 1 false alarm per 8 days), below the Phase 3 performance range. The model (Episave) uses an adjustable ensemble aggregation quantile parameter that allows trading sensitivity for false alarms. At one extreme, sensitivity can be raised to 100%. At the other extreme, the FPR can be reduced to 1/100 days at the cost of lower sensitivity (86%). Fine (2025) achieved 0.023/h FPR, within the Phase 3 performance range.
Three detection studies do not report FPR. Singh Rathore et al. (2024), Elemam et al. (2025), and Borujeny et al. (2013) lack standardized FPR reporting. This renders clinical assessment and comparison across studies impossible.

4.4.2 Forecasting Performance

Sensitivity in forecasting ranges from 51.2% to 82%. Individual study results are reported in Sections 4.2 and 4.3. Vieluf et al. (2025) achieved 82% sensitivity with 67% specificity, while Meisel et al. (2020) reported 51.2% sensitivity with LOSO validation.
Forecasting studies also report forecasting-specific metrics. Meisel et al. (2020) reported IoC of 14.1% and TiW of 43.7%. Nasseri et al. (2021) reported TiW ranging from 0.9 to 7.2 h/day. Stirling et al. (2021) reported mean prediction time of 37 min for hourly forecasts and 3 days for daily forecasts.
Forecasting AUC consistently falls between 0.74 and 0.75 across studies. This consistency suggests a performance ceiling for non-invasive forecasting approaches using wearable biosignals.

4.5 Detection, Forecasting, and Clinical Readiness

Detection and forecasting serve different clinical purposes. Detection aims to identify seizures as they occur for timely intervention. Forecasting aims to predict seizures before onset to allow preventive action. These different objectives lead to distinct technical approaches and validation requirements.

4.5.1 Clinical Objectives

Detection systems alert caregivers when a seizure occurs. This enables intervention during or immediately after the seizure. Detection is most valuable for generalized tonic-clonic or focal to bilateral tonic-clonic seizures where caregiver response can prevent injury or complications.
Forecasting systems provide advance warning of increased seizure likelihood. This allows patients to modify activities, take rescue medication, or avoid risky situations. Forecasting could improve quality of life by reducing the uncertainty of when seizures might occur.

4.5.2 Maturity Comparison

Detection shows greater clinical maturity than forecasting. Two detection studies achieve FPR performance comparable to Phase 3 studies in the ILAE systematic review. Commercial devices exist for detection including the Embrace watch with FDA 510(k) clearance and NightWatch with CE approval in Europe. No forecasting study meets a clear clinical benchmark. No device has regulatory approval for seizure forecasting.

Detection benefits from clearer clinical requirements. The ILAE systematic review of Phase 3 studies reported FPR of 0.2-0.67 per 24 hours (approximately 0.008-0.028/h) for validated GTCS/FBTCS detection devices (Beniczky et al., 2021). Forecasting lacks equivalent standardized guidelines. Forecasting AUC consistently falls between 0.74 and 0.80 across studies, suggesting a performance ceiling for non-invasive approaches, but the clinical acceptability of this performance level remains unclear.

Sample size disparities further reflect the maturity difference. Detection studies have a median sample size of 66 participants compared to 40 for forecasting studies. The largest detection study enrolled 384 patients across eight centers, while the largest forecasting study enrolled 70 patients.

4.5.3 Validation Phase Distribution

No study in this review has achieved Phase 3 prospective validation in home settings. Phase 3 validation requires prospective testing in the intended use environment with sufficient sample size and duration.

Two detection studies approach Phase 3 requirements. Spahr et al. (2025) conducted prospective multi-center validation with 384 patients. The study achieved FPR performance comparable to Phase 3 studies in the ILAE systematic review but was conducted in EMU settings rather than home. Dong et al. (2022) conducted prospective home monitoring with 68 patients but reported FPR above that performance range.

Fine (2025) represents Phase 1 validation with a small independent test set. The study achieved 100% sensitivity with 0.023/h FPR but tested only 10 seizures from 3 patients.

Forecasting studies remain in earlier validation phases. Vieluf et al. (2025) used Phase IV clinical trial data but employed retrospective analysis. Meisel et al. (2020) and Nasseri et al. (2021) represent Phase 2-equivalent validation with moderate samples.

4.5.4 Commercial Device Status

Six of 13 studies use commercially available devices. The Empatica E4 research wristband appears in three studies. The Embrace watch appears in one study. The NightWatch armband appears in one study. Fitbit smartwatches appear in one study.

The Embrace watch has FDA 510(k) clearance for seizure detection. NightWatch has CE approval in Europe. However, no device has regulatory approval specifically for seizure forecasting. Forecasting devices would need to establish clinical utility and safety through dedicated trials.

Seven studies use research prototypes without clear commercialization pathways. These prototypes demonstrate technical feasibility but may require substantial engineering for commercial deployment.

4.5.5 Real-World Deployment

Six studies include home or ambulatory validation. Dong et al. (2022) achieved the most extensive home validation with 788 overnight recordings over three months. Stirling et al. (2021) achieved the longest monitoring duration with a mean of 14.6 months per patient. Nasseri et al. (2021) achieved the most rigorous ambulatory validation with concurrent invasive EEG confirmation via the RNS system. However, only six patients were studied. Seven studies were conducted entirely in hospital or EMU settings. Hospital settings may not capture the diversity of real-world activities that cause false alarms. Exercise, cooking, household chores, and other activities present challenges not encountered in controlled hospital environments.

4.5.6 Deployment Architecture

Three studies report on-device or real-time deployment capability. Spahr et al. (2025) reported 112 ms inference time suitable for real-time processing. Borujeny et al. (2013) reported 316 mW power consumption for server-based processing. Dong et al. (2022) deployed on the NightWatch armband with real-time processing.

Cloud-based processing appears in three studies. Nasseri et al. (2021) and Ode et al. (2023) used cloud-based approaches. Cloud processing may introduce latency and connectivity issues. Offline processing appears in two studies without clear deployment pathways.

4.5.7 Barriers to Clinical Adoption

Several barriers impede clinical translation of wearable seizure monitoring. First, high FPR limits clinical utility. Six of eight detection studies report FPR above acceptable thresholds. Dong et al. (2022) reported 0.165/h FPR in home validation, Wang et al. (2025) reported 0.354/h, and Reintjes et al. (2025) reported FPR from 0.11/h to 65.62/h depending on method and configuration. High FPR causes alarm fatigue and may lead patients to abandon devices.

Second, limited seizure type coverage restricts applicability. Detection approaches focus on convulsive seizures with motor manifestations captured by accelerometer and gyroscope sensors. Focal aware seizures without motor signs remain undetectable with current wearable approaches. Non-convulsive seizures represent a substantial unmet need that may require invasive EEG monitoring.

Third, small sample sizes limit generalizability. Four studies have sample sizes below 20 patients. Singh Rathore et al. (2024) is a single-patient case study. Borujeny et al. (2013) studied only 3 patients. Small samples increase uncertainty about performance estimates and may not capture the full diversity of seizure presentations.

Fourth, validation setting affects real-world applicability. Seven studies were conducted entirely in hospital or EMU settings. Hospital environments may not capture the diversity of real-world activities that cause false alarms.

Fifth, device burden and battery life limit adherence. Multi-modal systems sampling multiple signals at high frequencies consume substantial power. Long-term monitoring requires daily charging which may reduce adherence over time.

4.5.8 Most Advanced Approaches

Spahr et al. (2025) represents the most clinically advanced detection approach. The study combines large sample size (384 patients), prospective multi-center design, excellent sensitivity (96%), and the lowest reported FPR among detection studies (0.0054/h). The ensemble 1D CNN algorithm operates in real-time with 112 ms inference time suitable for on-device processing. The main limitation is EMU setting rather than home validation.
Dong et al. (2022) represents the most advanced home-validated detection approach. The study achieved prospective home monitoring with 788 overnight recordings over three months using the NightWatch commercial device. FPR of 0.165/h exceeds the performance range reported in Phase 3 studies but may be acceptable for some patients prioritizing sensitivity over specificity.
Stirling et al. (2021) represents the most clinically advanced forecasting approach. The study achieved 100% patient success with long home validation using consumer Fitbit devices. The LSTM+RF+LR ensemble with weekly retraining adapts to individual patient patterns. However, forecasting lacks clear clinical benchmarks and regulatory pathways.

4.5.9 Reporting Practices

Detection and forecasting differ in metric reporting. Forecasting studies consistently report patient-level success rates. Detection studies rarely report individual patient outcomes. This reporting gap complicates assessment of which patients benefit from each approach.
Forecasting studies report forecasting-specific metrics including IoC and TiW. Detection studies emphasize sensitivity and FPR. These different metrics reflect different clinical priorities but complicate cross-domain comparison.
Key finding. Detection is closer to clinical translation than forecasting. Two detection approaches meet clinical FPR benchmarks. Forecasting shows consistent AUC around 0.75 but lacks clear clinical benchmarks and regulatory pathways.
Key limitation. No study has achieved all requirements for clinical deployment. All studies have limitations in sample size, validation setting, FPR, or generalizability. No study has achieved Phase 3 prospective validation in home settings with sufficient sample size and duration.
Limitation identified. Direct comparison between detection and forecasting is challenging due to different metrics, validation approaches, and clinical objectives. The field lacks standardized frameworks for assessing relative clinical value.

5 Discussion

This discussion synthesizes findings from 13 studies on wearable seizure detection and forecasting. The evidence reveals different maturity levels for detection versus forecasting, distinct technical requirements, and persistent barriers to clinical translation. We organize the discussion into four parts. First, we examine clinical implications regarding device readiness and coverage gaps. Second, we analyze technical trade-offs in modality selection, architecture design, and personalization strategies. Third, we identify limitations in the evidence base that constrain confidence in the findings. Fourth, we propose research priorities to advance the field toward clinical translation.

5.1 Clinical Implications

From a clinical perspective, three key findings emerge regarding the scope of seizure coverage, the readiness of detection devices, and the current state of forecasting technology.

5.1.1 Seizure Type Coverage Gap

Current wearable approaches address only a subset of seizure types. Accelerometer-based systems have shown to detect motor manifestations reliably and with acceptable sensitivity, but non-motor seizures remain difficult to detect. The ECG-based approach by Reintjes et al. (2025) examined whether autonomic signals could detect non-motor seizures. However, the observed sensitivity-FPR tradeoff prevents clinical interpretation. The lowest false alarm rate detected fewer than 3% of seizures, while configurations achieving acceptable sensitivity generated false alarm burdens orders of magnitude above clinically acceptable thresholds. This fundamental tradeoff demonstrates that single-lead ECG anomaly detection cannot currently provide clinically reliable seizure detection.

Forecasting approaches may theoretically address this gap by relying on autonomic modalities rather than movement sensors. However, the consistent AUC ceiling around 0.75 suggests fundamental limits in predicting seizures without EEG biomarkers. As Ayub et al. (2020) found, approximately 81% of seizures are focal onset and 19% are generalized tonic-clonic, meaning current accelerometer-based wearable approaches cannot detect the majority of seizures. This coverage gap represents a substantial unmet need that may require invasive monitoring or alternative biomarkers.

5.1.2 Detection Readiness

The detection literature demonstrates that clinically acceptable FPR is achievable with accelerometer-based approaches (see Section 4). Two studies achieve FPR performance comparable to Phase 3 studies in the ILAE systematic review, which reported 0.2-0.67 false alarms per 24 hours (approximately 0.008-0.028/h) (Beniczky et al., 2021). This suggests that motor seizure detection has reached sufficient maturity for clinical translation. However, this achievement remains concentrated in controlled EMU settings rather than real-world home environments.

The EMU-to-home translation gap represents a critical barrier. The two studies achieving benchmark FPR were validated in hospital settings where patient activities are restricted and controlled. Home environments introduce motion patterns from daily living that do not occur in EMUs, including exercise, household chores, and other routine activities. The substantial FPR increase observed in home validation (0.165/h versus 0.0054/h in EMU settings (Dong et al., 2026; Spahr et al., 2025)) suggests that algorithm robustness to real-world activities must become a research priority.

5.1.3 Forecasting Status

The forecasting literature shows a consistent performance ceiling around AUC 0.74 to 0.75 across different methods, modalities, and patient populations (see Section 4). This convergence suggests a fundamental limit on pre-ictal predictability using non-invasive wearable sensors. Unlike detection, where clear clinical benchmarks exist, forecasting lacks standardized validation guidelines. Without established performance targets, the clinical utility of AUC 0.75 remains uncertain.

The personalization strategy fundamentally affects forecasting utility. Global LOSO validation achieves only 43% patient success, meaning more than half of patients receive no benefit from population-based forecasting.

Patient-specific approaches achieve 100% success but require weeks to months of individual data collection before the system becomes useful. This cold-start problem represents a substantial barrier to clinical adoption, as newly diagnosed patients cannot wait months for protection.

No device has regulatory approval specifically for seizure forecasting. Forecasting devices would need to establish clinical utility and safety through dedicated trials. The lack of clear benchmarks complicates pathway to regulatory approval. In addition the current research remains sparse and unfocused with significantly different forecasting windows and methods.

5.2 Technical Insights and Trade-offs

These clinical observations reflect underlying technical trade-offs in modality selection, architecture design, and personalization strategy that shape what is achievable with current wearable approaches.

5.2.1 Modality Paradox

The evidence reveals a paradox regarding modality selection. Multi-modal approaches dominate the research field, with 69% of studies using multiple sensor types. However, the study achieving the lowest FPR uses single-modality accelerometry. This finding suggests that sensor fusion is not always necessary for reliable detection, and that biological specificity of motor manifestations may be more important than modality count.

The physiological requirements differ between detection and forecasting. Detection can succeed with movement sensors alone because ictal motor manifestations produce clear signals in ACC recordings. Forecasting requires autonomic modalities because pre-ictal changes must be detected before clinical symptoms appear. All forecasting studies incorporate EDA, ECG, or temperature data to capture these autonomic precursors. The contrast suggests that modality selection should be guided by the temporal phase of seizure activity being targeted.

5.2.2 Architecture Selection

Detection and forecasting applications employ different architectural patterns. Detection research shows methodological diversity including CNNs, LSTMs, ANNs, ensemble methods, and anomaly detection. This diversity may reflect the broader exploration phase of detection research. Forecasting studies concentrate heavily on LSTM variants, with three of four studies using this architecture. This convergence reflects the different temporal demands, as forecasting requires modeling longer-term dependencies than detection.

The persistence of competitive handcrafted feature approaches is noteworthy. Domain knowledge remains valuable despite the end-to-end learning trend, with engineered features achieving excellent performance in some studies. This suggests that purely data-driven approaches may not fully grasp the known physiological signatures of seizures or that current end-to-end approaches can be optimized further.

Ensemble methods achieve excellent results but require greater computational resources. The trade-off between ensemble complexity and deployability becomes important for commercial translation, as multi-model ensembles may be impractical for low-power wearable devices.

5.2.3 Personalization Dilemma

The personalization strategy presents a fundamental dilemma for clinical deployment. Global models enable immediate deployment without individual patient data, but only 43% of patients achieve performance above chance in LOSO validation. This means most patients receive no meaningful protection from population-based systems rendering them useless for clinical deployment.

Patient-specific approaches achieve 100% success rates but require substantial individual data collection. The cold-start problem means newly diagnosed patients must wait weeks or months before the system becomes useful. Mixed approaches combining population-level patterns with individual data may offer a middle path, but evidence remains limited to single studies.

5.2.4 The False Positive Challenge

The false positive challenge differs substantially between modalities. Accelerometer-based approaches can achieve clinically acceptable FPR below 0.05/h, but ECG-based approaches produce FPR an order of magnitude higher. Even the best ECG-based method exceeded the clinical benchmark by a substantial margin. This suggests that cardiac signals alone cannot reliably distinguish seizures from normal daily activities.

Multi-modal sensor fusion does not automatically solve the false positive problem. Studies combining multiple modalities still report FPR above clinical targets. The challenge may stem from the fundamental similarity between seizure-related physiological changes and those occurring during normal activities such as exercise, stress, or sleep transitions.

Forecasting faces a different performance ceiling. The consistency of AUC around 0.75 across different methods and modalities suggests fundamental limits on pre-ictal predictability using non-invasive sensors. This ceiling may reflect genuine biological variability in pre-ictal states that cannot be resolved with current wearable modalities and requires furhter research on the exact biological mechanisms during an epileptic seizure.

5.3 Limitations of the Evidence Base

The strength of these conclusions is limited by evidence quality across four dimensions: sample size constraints, validation setting bias, inconsistent metrics reporting, and insufficient monitoring durations.

5.3.1 Sample Size Constraints

The evidence base is constrained by small sample sizes in multiple studies. Five studies have sample sizes below 20 participants, including single-patient case studies and 3-patient evaluations. Small samples limit the reliability of performance estimates and reduce confidence that results will generalize to broader epilepsy populations especially since epilepsy is in itself drastically different between multiple patients. The forecasting literature is particularly affected, with studies limited to 6-11 patients due to invasive EEG confirmation requirements.

5.3.2 Validation Setting Bias

Eight of 13 studies were conducted in hospital or EMU settings, creating a systematic bias that may underestimate real-world FPR. Hospital environments lack the diversity of daily activities that generate false alarms in home settings as discussed in Section 5.1.2.

5.3.3 Reporting Inconsistency

Inconsistent metrics reporting hinders cross-study comparison. Forecasting studies routinely report patient-level success rates, enabling assessment of which patients benefit from the technology. Detection studies rarely provide this granularity, with only two of nine studies reporting individual patient outcomes. This reporting gap prevents assessment of whether detection algorithms work reliably for all patients or only a subset. High aggregate sensitivity may mask substantial variability across patients that remains invisible without patient-level reporting. This is probable since epileptic seizures vary widely between patients. The model has a strong likelihood to focus on the biggest subset of patients with easier to detect and more similar seizure patterns. This would result in poor performance for patients with irregular seizure patterns.

5.3.4 Monitoring Duration and Cold-Start Problems

Patient-specific approaches require weeks to months of individual data before becoming useful, creating a cold-start problem for newly diagnosed patients. This requirement conflicts with the clinical need for immediate protection after diagnosis. While global models enable immediate deployment, they achieve substantially lower patient success rates and possibly low generalizability. The trade-off between immediate deployment and individualized performance remains unresolved without evidence from hybrid approaches.

5.4 Future Directions

Building on the mentioned limitations, nine research priorities emerge to advance wearable seizure monitoring toward clinical utility.

First, forecasting needs standardized clinical benchmarks. Detection has established clinical benchmarks from the ILAE systematic review of Phase 3 studies, which reported FPR of 0.2-0.67 per 24 hours (approximately 0.008-0.028/h) for convulsive seizure detection (Beniczky et al., 2021). Forecasting lacks an equivalent standard. AUC values around 0.74 to 0.75 appear consistently across forecasting studies, but the clinical meaning of this performance level remains unclear. Metrics such as IoC and TiW need standardization. The field should establish minimum acceptable performance levels for forecasting devices to enable clinical translation.

Second, large-scale home validation is needed. Only two detection studies include prospective home validation. Hospital and EMU settings may not capture the diversity of real-world activities that cause false alarms. Exercise, cooking, household chores, and other daily activities present challenges not encountered in controlled environments. The 0.165/h FPR reported by Dong et al. (2022) in home settings exceeds the clinical benchmark however it does come close. Future studies should prioritize prospective validation in intended use environments with sufficient sample sizes and duration.

Third, novel modalities are needed for non-convulsive seizures. Current approaches detect only motor seizures with convulsive manifestations. Focal aware seizures without motor signs remain undetectable with current wearable approaches. Spahr et al. (2025) achieved excellent performance for convulsive seizures using accelerometer data alone, but this approach cannot detect seizures without movement. Multi-modal autonomic approaches show promise but have not addressed non-convulsive seizure types. Novel biosignals, advanced feature engineering or futher research into the exact mechanics of seizures may be required to detect the subtle physiological changes associated with focal seizures.

Fourth, adaptive personalization strategies could address the trade-off between deployability and individual performance. Global models enable immediate deployment but may not work equally well for all patients. Meisel et al. (2020) achieved 43% patient success with LOSO validation, indicating substantial inter-patient variability. Patient-specific models achieve higher individual success but require weeks to months of individual data. Stirling et al. (2021) required a mean of 14.6 months per patient. Nasseri et al. (2021) required 6+ months per patient (median 220 days). Mixed approaches that combine population-level patterns with individual adaptation, such as the method used by Vieluf et al. (2025), may offer a balance. Research should address the cold-start problem by developing models that can initialize from population data and adapt efficiently to individual patients.

Fifth, real-time processing and edge deployment require more attention. Most studies in this review ran algorithms offline on PCs or servers rather than on wearable devices. Clinical utility demands real-time detection and forecasting on-device to enable timely interventions and eliminating the risk posed by connection loss. This requires optimization for low power consumption and limited memory. Spahr et al. (2025) is an exception, achieving 112 ms inference time suitable for smartwatch integration. Their model (Episave) uses only 3D-ACC data, enabling integration with low-cost off-the-shelf smartwatches. This approach could expand accessibility to seizure monitoring for people with epilepsy in low- and middle-income countries where specialized devices are less available. Future work should report computational requirements and demonstrate feasible deployment on target hardware.

Sixth, model explainability and interpretability are needed for clinical adoption. Deep learning architectures such as CNNs, LSTMs, and autoencoders operate as black boxes with limited transparency. Clinicians and patients need to understand what features trigger detections or forecasts to trust these systems. Nasseri et al. (2021) performed ablation studies to examine feature contributions, and Reintjes et al. (2025) analyzed feature importance. However, systematic interpretability frameworks are missing. Future research should integrate explainable AI techniques and provide interpretable outputs for clinical decision-making.

Seventh, regulatory pathways and clinical integration need consideration. None of the reviewed studies address FDA or CE marking requirements for medical devices. Integration with clinical workflows, liability considerations for failed detections or forecasts, and ethical implications remain unexplored. Device approval pathways require rigorous validation data, standardized manufacturing processes, and post-market surveillance plans. Research should engage with regulatory frameworks early in development to facilitate clinical translation.

Eighth, algorithm robustness and generalizability require multi-site validation. Most studies used single-site or single-country data, limiting generalizability across populations and healthcare systems. Multi-continental validation with diverse demographic, geographic, and clinical characteristics is needed. Cross-device validation is also lacking. Studies typically evaluate algorithms on specific hardware such as Empatica E4, but performance may differ on consumer devices like Apple Watch or Fitbit. Standardized datasets and

benchmarks would enable fair comparison across approaches.
Ninth, long-term algorithm stability remains unaddressed. Seizure patterns evolve due to medication changes, disease progression, and physiological changes over time. Stirling et al. (2021) retrained models weekly, acknowledging performance drift. However, no study reports systematic evaluation of model validity over periods longer than several months. Longitudinal studies assessing how detection and forecasting performance decays over time are needed. Adaptive algorithms that can detect and compensate for performance degradation would improve clinical utility.

6 Conclusion

This systematic review analyzed 13 studies on wearable seizure detection and forecasting using deep learning approaches. The evidence base comprises 912 participants across nine detection studies and four forecasting studies published between 2013 and 2026.
The predominance of accelerometer-based detection reveals a fundamental constraint in wearable seizure monitoring. Six of nine detection studies rely on accelerometry, and the two studies achieving FPR rates comparable to ILAE Phase 3 both use ACC. Spahr et al. (2025) achieved 0.0054/h FPR with single-modality ACC. Fine (2025) achieved 0.023/h FPR with a 6-axis ACC and gyroscope band. This convergence on a single modality reflects the biological reality that convulsive seizures produce distinct motor signatures that deep learning can extract reliably. However, it also means current wearable approaches can serve only the subset of patients with motor manifestations. Approximately 81% of seizures are focal onset and 19% are generalized tonic-clonic (Ayub et al., 2020). While generalized tonic-clonic seizures are less common, they carry a 27-fold increased SUDEP risk compared to non-GTC seizures (Sveinsson et al., 2020), making their detection a clinical priority.
Forecasting studies face a different constraint. The consistent AUC plateau around 0.74 to 0.75 across studies using different methods, modalities, and patient populations suggests a fundamental limit on pre-ictal predictability using non-invasive sensors. Unlike detection, where the ictal motor event produces a clear signal, forecasting must rely on autonomic precursors that may be inherently less specific. The autonomic nervous system responds to stress, exercise, sleep transitions, and other daily events. These confounds may establish a performance ceiling for wearable forecasting that cannot be breached without EEG biomarkers or novel sensing modalities.
No study in this review has achieved all requirements for widespread clinical adoption. Commercial detection devices exist with FDA or CE approval, but the deep learning approaches reviewed here lack large-scale prospective validation. Forecasting faces an additional barrier. No device has regulatory approval for forecasting, and the field lacks established clinical benchmarks equivalent to the ILAE Phase 3 FPR standard. The performance achieved to date, while scientifically promising, may not be sufficient for regulatory approval or clinical utility.
The clinical pathway forward requires addressing three gaps. First, home validation at scale is needed to bridge the EMU-to-home translation gap. The two studies meeting FPR benchmarks were conducted in hospital settings. Real-world deployment introduces activities and confounds not captured in EMU environments. Second, forecasting needs standardized clinical benchmarks to define what level of AUC, IoC, or TiW constitutes clinically meaningful performance. Third, novel modalities or approaches are needed for focal seizures without motor manifestations. ECG-based detection shows some promise

for identifying autonomic changes during focal seizures, but current performance remains insufficient. Mixed personalization strategies that combine population patterns with individual adaptation may address the deployability versus performance trade-off without requiring months of individual data collection.

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